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Problem

- RNNs tie sequence positions to computation time, preventing within-example parallelism and slowing long-sequence training.
- Convolutions parallelize positions but need deeper paths to connect distant tokens.
- Can a sequence transducer model global dependencies using attention alone?

2

Architecture

STACKED ENCODER + DECODER

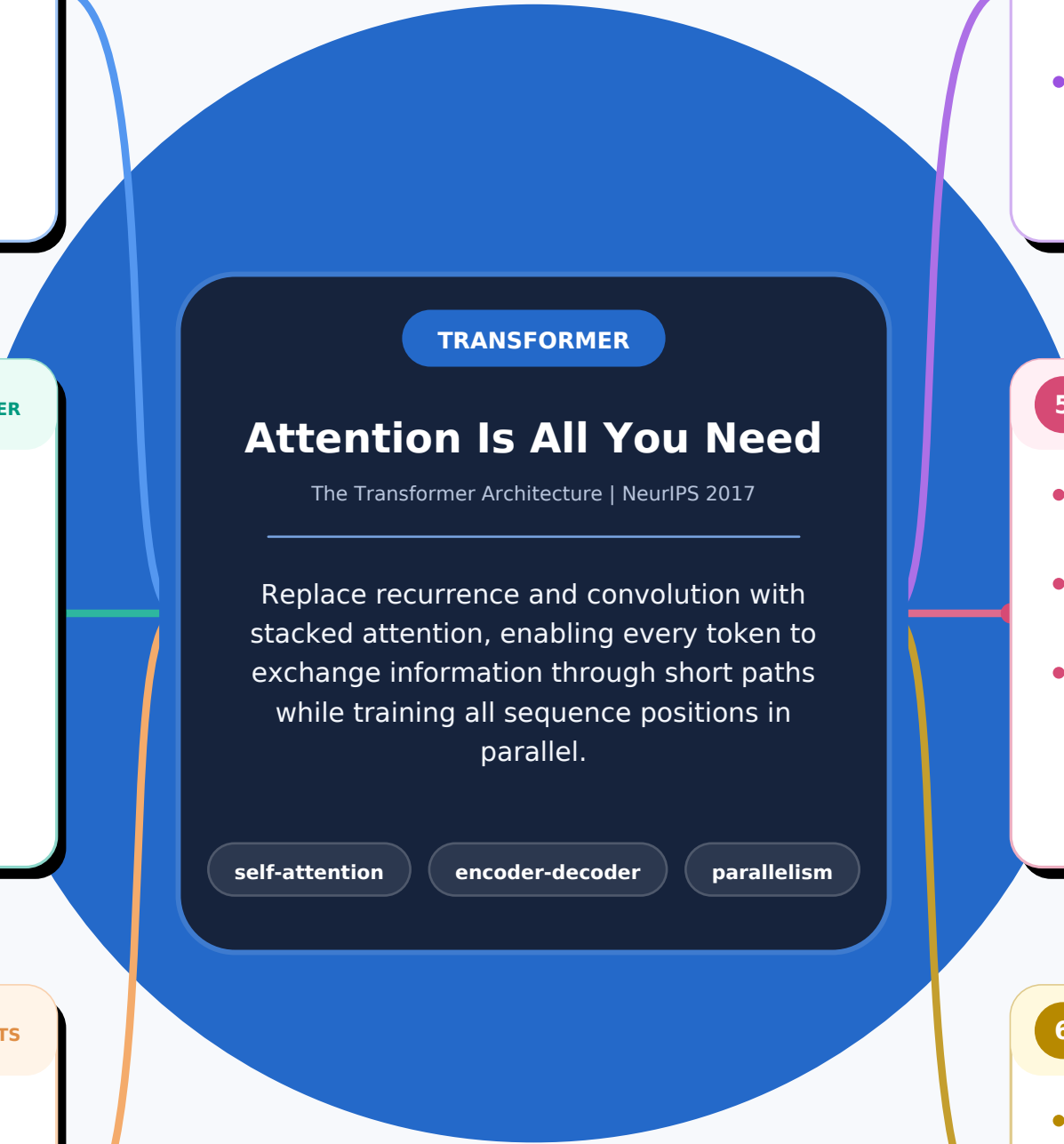
- The base model uses six encoder and six decoder layers with residual connections followed by LayerNorm.
- Each layer combines multi-head attention with a position-wise two-layer FFN; $d_{\text{model}}=512$ and $d_{\text{ff}}=2048$.
- Decoder self-attention is causally masked; encoder-decoder attention lets every output position read the entire input.

3

Attention

SCALED DOT PRODUCTS

- $\text{Attention}(Q,K,V) = \text{softmax}(Q K^T / \sqrt{d_k}) V$
- Scaling prevents large dot products from saturating softmax and shrinking gradients.
- Eight heads project Q , K and V into distinct 64-D subspaces, then concatenate diverse relational views.



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Order & efficiency

- Sinusoidal position encodings are added to token embeddings; learned positions perform similarly.
- Self-attention needs $O(n^2 d)$ work per layer but only $O(1)$ sequential operations and $O(1)$ maximum path length.
- For typical translation settings where n is smaller than d , it is faster than recurrent layers; restricted attention is proposed for long inputs.

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Evidence

- WMT14 English-to-German reaches 28.4 BLEU, over 2 BLEU above prior results including ensembles.
- English-to-French reaches 41.8 BLEU after 3.5 days on eight GPUs at a fraction of prior training cost.
- The same architecture generalizes to constituency parsing, reaching 92.7 F1 in the semi-supervised setting.

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Meaning & limits

A NEW SEQUENCE PRIMITIVE

- Attention exposes interpretable dependency patterns and makes training highly parallel, but autoregressive decoding remains sequential.
- Quadratic attention becomes costly for very long sequences and motivated later sparse/local variants.
- The paper established a general backbone later extended beyond text to images, audio, video and multimodal models.